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1 Miscellaneous

1.1 What to Look for

Wrong answer:
Print your solution! (Print debug output, as well.)
Are you clearing all DS between test cases?
Can your algorithm handle the whole range of input?
Read the full problem statement again.
Do you handle all corner cases correctly?
Have you understood the problem correctly?
Any uninitialized variables? Any overflows?
Confusing N and M, i and j, etc.?
Are you sure your algorithm works?
What special cases have you not thought of?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some testcases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Go for a small walk, e.g. to the toilet. Is your output format correct? (including ← whitespace)
Recode

Runtime error: In some judges, MLE is considered RTE (Check your memory use)
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Infinite recursion?
Any possible division by 0? Ex. mod 0
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits.

Force WA (asserts for RTE problems)

Time limit exceeded:
Do you have any possible infinite loops?
What is the complexity of your algorithm?
Are you copying a lot of unnecessary data? (References)
How big is the input and output? (consider scanf)
Avoid vector, map. (use arrays/unordered_map)
What do your teammates think about your algorithm?

Memory limit exceeded:
What is the max amount of memory your algorithm should need?
Are you clearing all DS between test cases?
Infinite recursion? (As in pushing into a DS infinitely)

1.2 Day of Date

```
// 0-based
const vector<int> T = {0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4}
int day(int d, int m, int y) {
    y -= (m < 3);
    return (y + y / 4 - y / 100 + y / 400 + T[m - 1] + d) % 7;
}
```

1.3 Number of Days since 1-1-1

```
int rdn(int d, int m, int y) {
    if(m < 3)
        --y, m += 12;
    return 365 * y + y / 4 - y / 100 + y / 400
        + (153 * m - 457) / 5 + d - 306;
}
```

1.4 Enumerate Subsets of a Bitmask

```
int x = 0;
do {
    // do stuff with the bitmask here
    x = (x + 1 + ~m) & m;
} while(x != 0);
```

1.5 Fast IO

```
int read() {
    char c;
    do {
        c = getchar_unlocked();
    } while(c < 33);
    int res = 0, mul = 1;
    if(c == '-') {
        mul = -1;
        c = getchar_unlocked();
    }
    for('0' <= c && c <= '9'; c = getchar_unlocked())
        res = res * 10 + c - '0';
    return res * mul;
}

void write(int x) {
    static char wbuf[10];
    if(x < 0) {
        putchar_unlocked('-');
        x = -x;
    }
    int idx = 0;
    for(; x; x /= 10)
        wbuf[idx++] = x % 10;
    if(idx == 0) putchar_unlocked('0');
    for(int i = idx - 1; i >= 0; --i) putchar_unlocked(wbuf[i] + '0');
}

void write(const char* s) {
    while(*s) {
        putchar_unlocked(*s);
        ++s;
    }
}
```

1.6 Josephus Problem

```
ll josephus(ll n, ll k) { // O(k log n)
    if(n == 1) return 0;
    if(k == 1) return n - 1;
    if(k > n) return (josephus(n - 1, k) + k) % n;
    ll cnt = n / k;
    ll res = josephus(n - cnt, k);
    res -= n % k;
    if(res < 0) res += n;
    else res += res / (k - 1);
    return res;
}

int josephus(int n, int k) { // O(n)
    int res = 0;
    for(int i = 1; i <= n; ++i) res = (res + k) % i;
    return res + 1;
}
```

1.7 Random Primes

36671 74101 724729 825827 924997 1500005681 2010408371 2010405347

1.8 RNG

```
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

// [mn, mx]
template<class T> T rand_int(T mn, T mx) {
    return uniform_int_distribution<T>(mn, mx)(rng);
}

// [mn,mx] NOT TYPO
template<class T> T rand_float(T mn, T mx) {
    return uniform_real_distribution<T>(mn, mx)(rng);
}
```

2 Data Structures

2.1 2D Segment Tree

```
struct Segtree2D {
    struct Segtree {
        struct node {
            int l, r, val;
            node* lc, * rc;
            node(int _l, int _r, int _val = INF) : l(_l), r(_r), val(_val),
                lc(NULL), rc(NULL) {}
        };
        typedef node* pnode;

        pnode root;

        Segtree(int l, int r) {
            root = new node(l, r);
        }

        void update(pnode& nw, int x, int val) {
            int l = nw->l, r = nw->r, mid = (l + r) / 2;
            if(l == r)
                nw->val = val;
            else {
                assert(l <= x && x <= r);
                pnode& child = x <= mid ? nw->lc : nw->rc;
                if(!child)
                    child = new node(x, x, val);
                else if(child->l <= x && x <= child->r)
                    update(child, x, val);
                else {
                    do {
                        if(x <= mid)
                            r = mid;
                        else
                            l = mid + 1;
                        mid = (l + r) / 2;
                    } while((x <= mid) == (child->l <= mid));
                    pnode nxt = new node(l, r);
                    if(child->l <= mid)
                        nxt->lc = child;
                    else
                        nxt->rc = child;
                    child = nxt;
                    update(nxt, x, val);
                }
            }
            nw->val = min(nw->lc ? nw->lc->val : INF,
                nw->rc ? nw->rc->val : INF);
        }

        int query(pnode& nw, int x1, int x2) {
            if(!nw)
                return INF;
            int& l = nw->l, &r = nw->r;
```

```

    if(r < x1 || x2 < l)
        return INF;
    if(x1 <= l && r <= x2)
        return nw->val;
    int ret = min(query(nw->lc, x1, x2),
                  query(nw->rc, x1, x2));
    return ret;
}

void update(int x, int val) {
    assert(root->l <= x && x <= root->r);
    update(root, x, val);
}

int query(int l, int r) {
    return query(root, l, r);
}
};

struct node {
    int l, r;
    Segtree y;
    node* lc, * rc;
    node(int _l, int _r) : l(_l), r(_r), y(0, MAX),
                        lc(NULL), rc(NULL) {}
};
typedef node* pnode;

pnode root;

Segtree2D(int l, int r) {
    root = new node(l, r);
}

void update(pnode& nw, int x, int y, int val) {
    int& l = nw->l, &r = nw->r, mid = (l + r) / 2;
    if(l == r)
        nw->y.update(y, val);
    else {
        if(x <= mid) {
            if(!nw->lc)
                nw->lc = new node(l, mid);
            update(nw->lc, x, y, val);
        } else {
            if(!nw->rc)
                nw->rc = new node(mid + 1, r);
            update(nw->rc, x, y, val);
        }
        val = min(nw->lc ? nw->lc->y.query(y, y) : INF,
                  nw->rc ? nw->rc->y.query(y, y) : INF);
        nw->y.update(y, val);
    }
}

int query(pnode& nw, int x1, int x2, int y1, int y2) {
    if(!nw)
        return INF;
    int& l = nw->l, &r = nw->r;
    if(r < x1 || x2 < l)
        return INF;
    if(x1 <= l && r <= x2)
        return nw->y.query(y1, y2);
    int ret = min(query(nw->lc, x1, x2, y1, y2),
                  query(nw->rc, x1, x2, y1, y2));
    return ret;
}

void update(int x, int y, int val) {
    assert(root->l <= x && x <= root->r);
    update(root, x, y, val);
}

```

```

int query(int x1, int x2, int y1, int y2) {
    return query(root, x1, x2, y1, y2);
}
};

```

2.2 Fenwick RU-RQ

```

void updtRL(int l, int r, ll val) {
    updt(BIT1, l, val), updt(BIT1, r + 1, -val);
    updt(BIT2, l, val * (l - 1)), updt(BIT2, r + 1, -val * r);
}
ll query(int k) {
    return que(BIT1, k) * k - que(BIT2, k);
}

```

2.3 Heavy-Light Decomposition

```

struct HLD {
    int n;
    vector<int> id, size, idx, up, root, st;
    vector<vector<int>> adj, chain;
    SegTree seg;

    HLD(const vector<vector<int>>& edges) :
        n(edges.size()), id(n, -1), size(n, -1), idx(n, -1),
        up(n, -1), adj(edges), seg(n) {
        precompute(0, -1);
        decompose(0, -1);
        int cnt = 0;
        st.resize(chain.size());
        for(int i = 0; i < (int) chain.size(); ++i) {
            st[i] = cnt;
            cnt += chain[i].size();
        }
    }

    void precompute(int pos, int dad) {
        size[pos] = 1;
        up[pos] = dad;
        for(auto& i : adj[pos]) {
            if(i != dad) {
                precompute(i, pos);
                size[pos] += size[i];
            }
        }
    }

    void decompose(int pos, int dad) {
        if(id[pos] == -1) {
            id[pos] = chain.size();
            root.push_back(pos);
            chain.emplace_back();
        }
        idx[pos] = chain[id[pos]].size();
        chain[id[pos]].push_back(pos);
        int mx = 0, heavy = -1;
        for(auto& i : adj[pos]) {
            if(i != dad && size[i] > mx) {
                mx = size[i];
                heavy = i;
            }
        }
        if(heavy != -1)
            id[heavy] = id[pos];
        for(auto& i : adj[pos]) {
            if(i != dad)
                decompose(i, pos);
        }
    }
}

```

```

}

void update(int ch, int l, int r, int val) {
    seg.update(st[ch] + l, st[ch] + r, val);
}

int query(int ch, int l, int r, int val) {
    return seg.query(st[ch] + l, st[ch] + r, val);
}
};

// how to move from u to v
while(1) {
    if(hld.id[u] == hld.id[v]) {
        if(hld.idx[u] > hld.idx[v])
            swap(u, v);
        hld.update(hld.id[u], hld.idx[u], hld.idx[v], w);
        // or hld.query(hld.id[u], hld.idx[u], hld.idx[v]);
        break;
    }
    if(hld.id[u] < hld.id[v])
        swap(u, v);
    hld.update(hld.id[u], 0, hld.idx[u], w);
    // or hld.query(hld.id[u], 0, hld.idx[u]);
    u = hld.up[hld.root[hld.id[u]]];
}

```

2.4 Li-Chao Tree

```

// max li-chao tree
// works for the range [0, MAX - 1]
// if min li-chao tree:
// replace every call to max() with min() and every > with <
// also replace -INF with INF

struct Func {
    ll m, c;
    ll operator()(ll x) {
        return x * m + c;
    }
};

const int MAX = 1e9 + 1;
const ll INF = 1e18;
const Func NIL = {0, -INF};

struct Node {
    Func f;
    Node* lc;
    Node* rc;

    Node() : f(NIL), lc(nullptr), rc(nullptr) {}
    Node(const Node& n) : f(n.f), lc(nullptr), rc(nullptr) {}
};

Node* root = new Node;

void insert(Func f, Node* cur = root, int l = 0, int r = MAX - 1) {
    int m = l + (r - l) / 2;
    bool left = f(l) > cur->f(l);
    bool mid = f(m) > cur->f(m);
    if(mid)
        swap(f, cur->f);
    if(l != r) {
        if(left != mid) {
            if(!cur->lc)
                cur->lc = new Node(*cur);
            insert(f, cur->lc, l, m);
        } else {
            if(!cur->rc)

```

```

        cur->rc = new Node(*cur);
        insert(f, cur->rc, m + 1, r);
    }
}

ll query(ll x, Node* cur = root, int l = 0, int r = MAX - 1) {
    if(!cur)
        return -INF;
    if(l == r)
        return cur->f(x);
    int m = l + (r - l) / 2;
    if(x <= m)
        return max(cur->f(x), query(x, cur->lc, l, m));
    else
        return max(cur->f(x), query(x, cur->rc, m + 1, r));
}

```

2.5 STL PBDS

```

// ost = ordered set
// omp = ordered map
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;

template<class T>
using ost = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
template<class T, class U>
using omp = tree<T, U, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

```

2.6 Treap

```

// Complexity: O(log N) for split and merge
//
// empty treap: Treap* tr = nullptr;
// insert v at x: [l, r] = split(tr, x), m = Treap(v), merge lmr
// delete at x: [l, r] = split(tr, x), [m, r] = split(r, 1), merge lr
// lazy prop: propagate every time a node is accessed

mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

using Key = int;

struct Treap {
    Key val;
    Treap* left;
    Treap* right;
    int prio, sz;
    Treap() {}
    Treap(int _val);
};

int size(Treap* tr) {
    return tr ? tr->sz : 0;
}

void update(Treap* tr) {
    tr->sz = 1 + size(tr->left) + size(tr->right);
}

Treap::Treap(Key _val) :
    val(_val), left(nullptr), right(nullptr), prio(rng()) {
    update(this);
}

```

```

struct Node {
    Np p, c[2];    // parent, children
    bool flip = 0; // subtree flipped or not
    int vtx;
    ll val, sz;    // value in node, # nodes in current splay tree
    int sub, vsub = 0; // # of nodes in CC
    ll stsum;      // sum of all val in the splay tree
    // after access(): from original root to this node
    Node(int _vtx, int _val = 1) : vtx(_vtx), val(_val) {
        p = c[0] = c[1] = nullptr;
        calc();
    }

    friend int getSz(Np x) {
        return x ? x->sz : 0;
    }
    friend int getSub(Np x) {
        return x ? x->sub : 0;
    }
    friend ll getStSum(Np x) {
        return x ? x->stsum : 0;
    }

    void prop() { // lazy prop
        if(!flip)
            return;
        swap(c[0], c[1]);
        flip = 0;
        for(int i = 0; i < 2; i++) {
            if(c[i])
                c[i]->flip ^= 1;
        }
    }
    void calc() { // recalc vals
        for(int i = 0; i < 2; i++) {
            if(c[i])
                c[i]->prop();
        }
        sz = 1 + getSz(c[0]) + getSz(c[1]);
        sub = 1 + getSub(c[0]) + getSub(c[1]) + vsub;
        stsum = val + getStSum(c[0]) + getStSum(c[1]);
    }

    // SPLAY TREE OPERATIONS
    int dir() { // p is path-parent pointer
        if(!p)
            return -2;
        for(int i = 0; i < 2; i++) {
            if(p->c[i] == this)
                return i;
        }
        return -1; // -> not in current splay tree
    }
    // test if root of current splay tree
    bool isRoot() {
        return dir() < 0;
    }
    friend void setLink(Np x, Np y, int d) { // x is orig parent
        if(y)
            y->p = x;
        if(d >= 0)
            x->c[d] = y;
    }
    void rot() { // assume p and p->p propagated
        assert(!isRoot());
        int x = dir();
        Np pa = p;
        setLink(pa->p, this, pa->dir());
        setLink(pa, c[x ^ 1], x);
        setLink(this, pa, x ^ 1);
        pa->calc();
    }
}

```

```

pair<Treap*, Treap*> split(Treap* tr, int sz) {
    if(!tr) return {nullptr, nullptr};
    int left_sz = size(tr->left);
    if(sz <= left_sz) {
        auto [left, mid] = split(tr->left, sz);
        tr->left = mid;
        update(tr);
        return {left, tr};
    } else {
        auto [mid, right] = split(tr->right, sz - left_sz - 1);
        tr->right = mid;
        update(tr);
        return {tr, right};
    }
}

Treap* merge(Treap* l, Treap* r) {
    if(!l)
        return r;
    if(!r)
        return l;
    if(l->prio < r->prio) {
        l->right = merge(l->right, r);
        update(l);
        return l;
    } else {
        r->left = merge(l, r->left);
        update(r);
        return r;
    }
}

```

2.7 Unordered Map Custom Hash

```

struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

unordered_map<int, int, custom_hash> umap;

```

2.8 Mo's on Tree

$ST(u) \leq ST(v)$
 $P = LCA(u, v)$
 If $P = u$, query $[ST(u), ST(v)]$
 Else query $[EN(u), ST(v)] + [ST(P), ST(P)]$

2.9 Link-Cut Tree

```

// sz for path queries
// sub, vsub for subtree queries
// x->access() brings x to the top and propagates it,
// x's left subtree is path from x to root
// after access, sub is no. of nodes in CC of x,
// vsub is no. of nodes under x
// use makeRoot() for arbitrary path queries

typedef struct Node* Np;

```

```

}
void splay() { // bring this node to the root of splay tree
    while(!isRoot() && !p->isRoot()) {
        p->p->prop(), p->prop(), prop();
        dir() == p->dir() ? p->rot() : rot();
        rot();
    }
    if(!isRoot())
        p->prop(), prop(), rot();
    prop();
    calc();
}
Np fbo(int b) { // find by order
    prop();
    int z = getSz(c[0]); // of splay tree
    if(b == z) {
        splay();
        return this;
    }
    return b < z ? c[0]->fbo(b) : c[1]->fbo(b - z - 1);
}

// BASE OPERATIONS
// bring this to top of splay tree (not impacting original tree)
void access() {
    for(Np v = this, pre = nullptr; v; v = v->p) {
        v->splay(); // now switch virtual children
        if(pre)
            v->vsub -= pre->sub;
        if(v->c[1])
            v->vsub += v->c[1]->sub;
        v->c[1] = pre;
        v->calc();
        pre = v;
    }
    splay();
    assert(!c[1]); // right subtree is empty
}
void makeRoot() { // of the original tree
    access();
    flip ^= 1;
    access();
    assert(!c[0] && !c[1]);
}

// QUERIES
friend Np lca(Np x, Np y) {
    if(x == y)
        return x;
    x->access(), y->access();
    if(!x->p)
        return nullptr;
    x->splay();
    return x->p ? x : y; // y was below x in latter case
} // access at y did not affect x -> not connected
friend bool connected(Np x, Np y) {
    return lca(x, y);
}

// no. of nodes above; distance to root in original tree
int distRoot() {
    access();
    return getSz(c[0]);
}
}
Np getRoot() { // get root of LCT component in original tree
    access();
    Np a = this;
    while(a->c[0])
        a = a->c[0], a->prop();
    a->access();
    return a;
}

```

```

}
Np getPar(int b) { // get b-th parent on path to root
    access();
    b = getSz(c[0]) - b;
    assert(b >= 0);
    return fbo(b);
} // can also get min, max on path to root, etc

// MODIFICATIONS
void setVal(int v) {
    access();
    val = v;
    calc();
}
void addVal(int v) {
    access();
    val += v;
    calc();
}
}
friend void link(Np x, Np y, bool force = 1) {
    assert(!connected(x, y));
    if(force) {
        y->makeRoot(); /// make x par of y; x -> y
    } else {
        y->access();
        assert(!y->c[0]);
    }
    x->access();
    setLink(y, x, 0);
    y->calc();
}
friend void cut(Np y) { // cut y from its parent
    y->access();
    assert(y->c[0]);
    y->c[0]->p = NULL;
    y->c[0] = NULL;
    y->calc();
}
friend void cut(Np x, Np y) { // if x, y adj in tree
    x->makeRoot();
    y->access();
    assert(y->c[0] == x && !x->c[0] && !x->c[1]);
    cut(y);
}
};
Np LCT[maxV];

```

3 Dynamic Programming

3.1 DP Convex Hull

```

/* dp[i] = min k<i {dp[k] + x[i]*m[k]}
   Make sure gradient (m[i]) is either non-increasing if min,
   or non-decreasing if max. x[i] must be non-decreasing. just sort */
int y[N], m[N];
// while this is true, pop back from dq. a=new line, b=last, c=2nd last
bool cekx(int a, int b, int c) {
    // if not enough, change to cross mul
    // if cross mul, beware of negative denominator, and overflow
    return (double)(y[b] - y[a]) / (m[a] - m[b]) <= (double)(y[c] - y[b]) /
           (m[b] - m[c]);
}

```

3.2 DP DNC

```

void f(int rem, int l, int r, int optl, int optr) {
    if(l > r)
        return;
}

```

```
int mid = l + r >> 1;
int opt = MOD, optid = mid;
for(int i = optl; i <= mid && i <= optr; ++i) {
    if(dp[rem - 1][i] + c[i][mid] < opt) {
        opt = dp[rem - 1][i] + c[i][mid];
        optid = i;
    }
}
dp[rem][mid] = opt;
f(rem, l, mid - 1, optl, optid);
f(rem, mid + 1, r, optid, optr);
return;
}
rep(i, 1, n)dp[1][i] = c[0][i];
rep(i, 2, k)f(i, i, n, i, n);
```

3.3 DP Knuth-Yao

```
// opt[i+1][j] <= opt[i][j] <= opt[i][j+1]
// dp[i][j] = min{k} dp[i][k]+dp[k][j]+cost[i][j]
for(int k = 0; k <= n; k++) {
    for(int i = 0; i + k <= n; i++) {
        if(k < 2)
            dp[i][i + k] = 0, opt[i][i + k] = i;
        else {
            int sta = opt[i][i + k - 1];
            int end = opt[i + 1][i + k];
            for(int j = sta; j <= end; j++) {
                if(dp[i][j] + dp[j][i + k] + cost[i][i + k] < dp[i][i + k]) {
                    dp[i][i + k] = dp[i][j] + dp[j][i + k] + cost[i][i + k];
                    opt[i][i + k] = j;
                }
            }
        }
    }
}
```

4 Geometry

4.1 Geometry Template

```
/*
TABLE OF CONTENT
0. Basic Rule
    0.1. Everything is in double
    0.2. Every comparison use EPS
    0.3. Every degree in rad
1. General Double Operation
    1.1. const double EPS=1E-9
    1.2. const double PI=acos(-1.0)
    1.3. const double INF=1E9
    1.3. between_d(double x,double l,double r)
        check whether x is between l and r inclusive with EPS
    1.4. same_d(double x,double y)
        check whether x=y with EPS
    1.5. dabs(double x)
        absolute value of x
2. Point
    2.1. struct point
        2.1.1. double x,y
            cartesian coordinate of the point
        2.1.2. point()
            default constructor
        2.1.3. point(double _x,double _y)
            constructor, set the point to (_x,_y)
        2.1.4. bool operator< (point other)
            regular pair<double,double> operator < with EPS
        2.1.5. bool operator==(point other)
```

```
regular pair<double,double> operator == with EPS
2.2. hypot(point P)
    length of hypotenuse of point P to (0,0)
2.3. e_dist(point P1,point P2)
    euclidean distance from P1 to P2
2.4. m_dist(point P1,point P2)
    manhattan distance from P1 to P2
2.5. point rotate(point P,point O,double angle)
    rotate point P from the origin O by angle ccw
3. Vector
    3.1. struct vec
        3.1.1. double x,y
            x and y magnitude of the vector
        3.1.2. vec()
            default constructor
        3.1.3. vec(double _x,double _y)
            constructor, set the vector to (_x,_y)
        3.1.4. vec(point A,point B)
            constructor, set the vector to vector AB (A->B)
*/
/*General Double Operation*/

const double PI = acos(-1.0);
const double INF = 1E9;
double between_d(double x, double l, double r) {
    return (min(l, r) <= x + EPS && x <= max(l, r) + EPS);
}
double same_d(double x, double y) {
    return between_d(x, y, y);
}
double dabs(double x) {
    if(x < EPS)
        return -x;
    return x;
}
/*Point*/
struct point {
    double x, y;
    point() {
        x = y = 0.0;
    }
    point(double _x, double _y) {
        x = _x;
        y = _y;
    }
    bool operator< (point other) {
        if(x < other.x + EPS)
            return true;
        if(x + EPS > other.x)
            return false;
        return y < other.y + EPS;
    }
    bool operator==(point other) {
        return same_d(x, other.x) && same_d(y, other.y);
    }
};
double e_dist(point P1, point P2) {
    return hypot(P1.x - P2.x, P1.y - P2.y);
}
double m_dist(point P1, point P2) {
    return dabs(P1.x - P2.x) + dabs(P1.y - P2.y);
}
double pointBetween(point P, point L, point R) {
    return (e_dist(L, P) + e_dist(P, R) == e_dist(L, R));
}
bool collinear(point P, point L,
    point R) { //newly added(luis), cek 3 poin segaris
    return P.x * (L.y - R.y) + L.x * (R.y - P.y) + R.x * (P.y - L.y) ==
        0; // bole gnti "dabs(x)<"EPS
}
```

```

/*Vector*/
struct vec {
    double x, y;
    vec() {
        x = y = 0.0;
    }
    vec(double _x, double _y) {
        x = _x;
        y = _y;
    }
    vec(point A) {
        x = A.x;
        y = A.y;
    }
    vec(point A, point B) {
        x = B.x - A.x;
        y = B.y - A.y;
    }
};
vec scale(vec v, double s) {
    return vec(v.x * s, v.y * s);
}
vec flip(vec v) {
    return vec(-v.x, -v.y);
}
double dot(vec u, vec v) {
    return (u.x * v.x + u.y * v.y);
}
double cross(vec u, vec v) {
    return (u.x * v.y - u.y * v.x);
}
double norm_sq(vec v) {
    return (v.x * v.x + v.y * v.y);
}
point translate(point P, vec v) {
    return point(P.x + v.x, P.y + v.y);
}
point rotate(point P, point O, double angle) {
    vec v(O);
    P = translate(P, flip(v));
    return translate(point(P.x * cos(angle) - P.y * sin(angle),
        P.x * sin(angle) + P.y * cos(angle)), v);
}
point mid(point P, point Q) {
    return point((P.x + Q.x) / 2, (P.y + Q.y) / 2);
}
double angle(point A, point O, point B) {
    vec OA(O, A), OB(O, B);
    return acos(dot(OA, OB) / sqrt(norm_sq(OA) * norm_sq(OB)));
}
int orientation(point P, point Q, point R) {
    vec PQ(P, Q), PR(P, R);
    double c = cross(PQ, PR);
    if(c < -EPS)
        return -1;
    if(c > EPS)
        return 1;
    return 0;
}
/*Line*/
struct line {
    double a, b, c;
    line() {
        a = b = c = 0.0;
    }
    line(double _a, double _b, double _c) {
        a = _a;
        b = _b;
        c = _c;
    }
    line(point P1, point P2) {

```

```

        if(P1 < P2)
            swap(P1, P2);
        if(same_d(P1.x, P2.x))
            a = 1.0, b = 0.0, c = -P1.x;
        else
            a = -(P1.y - P2.y) / (P1.x - P2.x), b = 1.0, c = -(a * P1.x) - P1.y;
    }
    line(point P, double slope) {
        if(same_d(slope, INFD))
            a = 1.0, b = 0.0, c = -P.x;
        else
            a = -slope, b = 1.0, c = -(a * P.x) - P.y;
    }
    bool operator==(line other) {
        return same_d(a, other.a) && same_d(b, other.b) && same_d(c, other.c);
    }
    double slope() {
        if(same_d(b, 0.0))
            return INFD;
        return -(a / b);
    }
};
bool paralel(line L1, line L2) {
    return same_d(L1.a, L2.a) && same_d(L1.b, L2.b);
}
bool intersection(line L1, line L2, point& P) {
    if(paralel(L1, L2))
        return false;
    P.x = (L2.b * L1.c - L1.b * L2.c) / (L2.a * L1.b - L1.a * L2.b);
    if(same_d(L1.b, 0.0))
        P.y = -(L2.a * P.x + L2.c);
    else
        P.y = -(L1.a * P.x + L1.c);
    return true;
}
double pointToLine(point P, point A, point B, point& C) {
    vec AP(A, P), AB(A, B);
    double u = dot(AP, AB) / norm_sq(AB);
    C = translate(A, scale(AB, u));
    return e_dist(P, C);
}
double lineToLine(line L1, line L2) {
    if(!paralel(L1, L2))
        return 0.0;
    return dabs(L2.c - L1.c) / sqrt(L1.a * L1.a + L1.b * L1.b);
}
/*Line Segment*/
struct segment {
    point P, Q;
    line L;
    segment() {
        point T1;
        P = Q = T1;
        line T2;
        L = T2;
    }
    segment(point _P, point _Q) {
        P = _P;
        Q = _Q;
        if(Q < P)
            swap(P, Q);
        line T(P, Q);
        L = T;
    }
    bool operator==(segment other) {
        return P == other.P && Q == other.Q;
    }
};
bool onSegment(point P, segment S) {
    if(orientation(S.P, S.Q, P) != 0)
        return false;

```



```

    return between_d(P.x, S.P.x, S.Q.x) && between_d(P.y, S.P.y, S.Q.y);
}
bool s_intersection(segment S1, segment S2) {
    double o1 = orientation(S1.P, S1.Q, S2.P);
    double o2 = orientation(S1.P, S1.Q, S2.Q);
    double o3 = orientation(S2.P, S2.Q, S1.P);
    double o4 = orientation(S2.P, S2.Q, S1.Q);
    if(o1 != o2 && o3 != o4)
        return true;
    if(o1 == 0 && onSegment(S2.P, S1))
        return true;
    if(o2 == 0 && onSegment(S2.Q, S1))
        return true;
    if(o3 == 0 && onSegment(S1.P, S2))
        return true;
    if(o4 == 0 && onSegment(S1.Q, S2))
        return true;
    return false;
}
double pointToSegment(point P, point A, point B, point& C) {
    vec AP(A, P), AB(A, B);
    double u = dot(AP, AB) / norm_sq(AB);
    if(u < EPS) {
        C = A;
        return e_dist(P, A);
    }
    if(u + EPS > 1.0) {
        C = B;
        return e_dist(P, B);
    }
    return pointToLine(P, A, B, C);
}
double segmentToSegment(segment S1, segment S2) {
    if(s_intersection(S1, S2))
        return 0.0;
    double ret = INFD;
    point dummy;
    ret = min(ret, pointToSegment(S1.P, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S1.Q, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S2.P, S1.P, S1.Q, dummy));
    ret = min(ret, pointToSegment(S2.Q, S1.P, S1.Q, dummy));
    return ret;
}
/*Circle*/
struct circle {
    point P;
    double r;
    circle() {
        point P1;
        P = P1;
        r = 0.0;
    }
    circle(point _P, double _r) {
        P = _P;
        r = _r;
    }
    circle(point P1, point P2) {
        P = mid(P1, P2);
        r = e_dist(P, P1);
    }
    circle(point P1, point P2, point P3) {
        vector<point> T;
        T.clear();
        T.pb(P1);
        T.pb(P2);
        T.pb(P3);
        sort(T.begin(), T.end());
        P1 = T[0];
        P2 = T[1];
        P3 = T[2];
        point M1, M2;

```

```

        M1 = mid(P1, P2);
        M2 = mid(P2, P3);
        point Q2, Q3;
        Q2 = rotate(P2, P1, PI / 2);
        Q3 = rotate(P3, P2, PI / 2);
        vec P1Q2(P1, Q2), P2Q3(P2, Q3);
        point M3, M4;
        M3 = translate(M1, P1Q2);
        M4 = translate(M2, P2Q3);
        line L1(M1, M3), L2(M2, M4);
        intersection(L1, L2, P);
        r = e_dist(P, P1);
    }
    bool operator==(circle other) {
        return (P == other.P && same_d(r, other.r));
    }
};
bool insideCircle(point P, circle C) {
    return e_dist(P, C.P) <= C.r + EPS;
}
bool c_intersection(circle C1, circle C2, point& P1, point& P2) {
    double d = e_dist(C1.P, C2.P);
    if(d > C1.r + C2.r) {
        return false; //d+EPS kalo butuh
    }
    if(d < dabs(C1.r - C2.r) + EPS)
        return false;
    double x1 = C1.P.x, y1 = C1.P.y, r1 = C1.r, x2 = C2.P.x, y2 = C2.P.y, r2 = C2.r;
    double a = (r1 * r1 - r2 * r2 + d * d) / (2 * d), h = sqrt(r1 * r1 - a * a);
    point T(x1 + a * (x2 - x1) / d, y1 + a * (y2 - y1) / d);
    P1 = point(T.x - h * (y2 - y1) / d, T.y + h * (x2 - x1) / d);
    P2 = point(T.x + h * (y2 - y1) / d, T.y - h * (x2 - x1) / d);
    return true;
}
bool lc_intersection(line L, circle C, point& P1, point& P2) {
    double a = L.a, b = L.b, c = L.c, x = 0.P.x, y = 0.P.y, r = 0.r;
    double A = a * a + b * b, B = 2 * a * b * y - 2 * a * c - 2 * b * b * x,
        C = b * b * x * x + b * b * y * y - 2 * b * c * y + c * c - b * b * r * r;
    double D = B * B - 4 * A * C;
    point T1, T2;
    if(same_d(b, 0.0)) {
        T1.x = c / a;
        if(dabs(x - T1.x) + EPS > r)
            return false;
        if(same_d(T1.x - r - x, 0.0) || same_d(T1.x + r - x, 0.0)) {
            P1 = P2 = point(T1.x, y);
            return true;
        }
        double dx = dabs(T1.x - x), dy = sqrt(r * r - dx * dx);
        P1 = point(T1.x, y - dy);
        P2 = point(T1.x, y + dy);
        return true;
    }
    if(same_d(D, 0.0)) {
        T1.x = -B / (2 * A);
        T1.y = (c - a * T1.x) / b;
        P1 = P2 = T1;
        return true;
    }
    if(D < EPS)
        return false;
    D = sqrt(D);
    T1.x = (-B - D) / (2 * A);
    T1.y = (c - a * T1.x) / b;
    P1 = T1;
    T2.x = (-B + D) / (2 * A);
    T2.y = (c - a * T2.x) / b;
    P2 = T2;
    return true;
}
bool sc_intersection(segment S, circle C, point& P1, point& P2) {

```

```

bool cek = lc_intersection(S.L, C, P1, P2);
if(!cek)
    return false;
double x1 = S.P.x, y1 = S.P.y, x2 = S.Q.x, y2 = S.Q.y;
bool b1 = between_d(P1.x, x1, x2) && between_d(P1.y, y1, y2);
bool b2 = between_d(P2.x, x1, x2) && between_d(P2.y, y1, y2);
if(P1 == P2)
    return b1;
if(b1 || b2) {
    if(!b1)
        P1 = P2;
    if(!b2)
        P2 = P1;
    return true;
}
return false;
}
/*Triangle*/
double t_perimeter(point A, point B, point C) {
    return e_dist(A, B) + e_dist(B, C) + e_dist(C, A);
}
double t_area(point A, point B, point C) {
    double s = t_perimeter(A, B, C) / 2;
    double ab = e_dist(A, B), bc = e_dist(B, C), ac = e_dist(C, A);
    return sqrt(s * (s - ab) * (s - bc) * (s - ac));
}
circle t_inCircle(point A, point B, point C) {
    vector<point> T;
    T.clear();
    T.pb(A);
    T.pb(B);
    T.pb(C);
    sort(T.begin(), T.end());
    A = T[0];
    B = T[1];
    C = T[2];
    double r = t_area(A, B, C) / (t_perimeter(A, B, C) / 2);
    double ratio = e_dist(A, B) / e_dist(A, C);
    vec BC(B, C);
    BC = scale(BC, ratio / (1 + ratio));
    point P;
    P = translate(B, BC);
    line AP1(A, P);
    ratio = e_dist(B, A) / e_dist(B, C);
    vec AC(A, C);
    AC = scale(AC, ratio / (1 + ratio));
    P = translate(A, AC);
    line BP2(B, P);
    intersection(AP1, BP2, P);
    return circle(P, r);
}
circle t_outCircle(point A, point B, point C) {
    return circle(A, B, C);
}
/*Polygon*/
struct polygon {
    vector<point> P;
    polygon() {
        P.clear();
    }
    polygon(vector<point>& _P) {
        P = _P;
    }
};
bool rayCast(point P, polygon& A) {
    point Q(P.x, 10000);
    line cast(P, Q);
    int cnt = 0;
    FOR(i, (int)(A.P.size()) - 1) {
        line temp(A.P[i], A.P[i + 1]);
        point I;

```

```

bool B = intersection(cast, temp, I);
if(!B)
    continue;
else if(I == A.P[i] || I == A.P[i + 1])
    continue;
else if(pointBetween(I, A.P[i], A.P[i + 1]) && pointBetween(I, P, Q))
    cnt++;
}
return cnt % 2 == 1;
}
// line segment p-q intersect with line A-B.
point lineIntersectSeg(point p, point q, point A, point B) {
    double a = B.y - A.y;
    double b = A.x - B.x;
    double c = B.x * A.y - A.x * B.y;
    double u = fabs(a * p.x + b * p.y + c);
    double v = fabs(a * q.x + b * q.y + c);
    return point((p.x * v + q.x * u) / (u + v), (p.y * v + q.y * u) / (u + v));
}
// cuts polygon Q along the line formed by point a -> point b
// (note: the last point must be the same as the first point)
vector<point> cutPolygon(point a, point b, const vector<point>& Q) {
    vector<point> P;
    for(int i = 0; i < (int)Q.size(); i++) {
        double left1 = cross(toVec(a, b), toVec(a, Q[i]));
        double left2 = 0;
        if(i != (int)Q.size() - 1)
            left2 = cross(toVec(a, b), toVec(a, Q[i + 1]));
        if(left1 > -EPS)
            P.push_back(Q[i]);
        if(left1 * left2 < -EPS)
            P.push_back(lineIntersectSeg(Q[i], Q[i + 1], a, b));
    }
    if(!P.empty() && !(P.back() == P.front()))
        P.push_back(P.front());
    return P;
}
circle minCoverCircle(polygon& A) {
    vector<point> p = A.P;
    point c;
    circle ret;
    double cr = 0.0;
    int i, j, k;
    c = p[0];
    for(i = 1; i < p.size(); i++) {
        if(e_dist(p[i], c) >= cr + EPS) {
            c = p[i], cr = 0;
            ret = circle(c, cr);
            for(j = 0; j < i; j++) {
                if(e_dist(p[j], c) >= cr + EPS) {
                    c = mid(p[i], p[j]);
                    cr = e_dist(p[i], c);
                    ret = circle(c, cr);
                    for(k = 0; k < j; k++) {
                        if(e_dist(p[k], c) >= cr + EPS) {
                            ret = circle(p[i], p[j], p[k]);
                            c = ret.P;
                            cr = ret.r;
                        }
                    }
                }
            }
        }
    }
    return ret;
}
/*Geometry Algorithm*/
double DP[110][110];
double minCostPolygonTriangulation(polygon& A) {
    if(A.P.size() < 3)
        return 0;

```

```
FOR(i, A.P.size()) {
    for(int j = 0, k = i; k < A.P.size(); j++, k++) {
        if(k < j + 2)
            DP[j][k] = 0.0;
        else {
            DP[j][k] = INFD;
            REP(l, j + 1, k - 1) {
                double cost = e_dist(A.P[j], A.P[k]) + e_dist(A.P[k], A.P[l]) + e_dist(A.P[l]↔
                    ],
                        A.P[j]);
                DP[j][k] = min(DP[j][k], DP[j][l] + DP[l][k] + cost);
            }
        }
    }
}
return DP[0][A.P.size() - 1];
}
```

4.2 Convex Hull

```
typedef double TD;                // for precision shifts
namespace GEOM {
    typedef pair<TD, TD> Pt;       // vector and points
    const TD EPS = 1e-9;
    const TD maxD = 1e9;
    TD cross(Pt a, Pt b, Pt c) {   // right hand rule
        TD v1 = a.first - c.first; // (a-c) X (b-c)
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u2 - v2 * u1;
    }
    TD cross(Pt a, Pt b) {          // a X b
        return a.first * b.second - a.second * b.first;
    }
    TD dot(Pt a, Pt b, Pt c) {      // (a-c) . (b-c)
        TD v1 = a.first - c.first;
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u1 + v2 * u2;
    }
    TD dot(Pt a, Pt b) {            // a . b
        return a.first * b.first + a.second * b.second;
    }
    TD dist(Pt a, Pt b) {
        return sqrt((a.first - b.first) * (a.first - b.first) +
            (a.second - b.second) * (a.second - b.second));
    }
    TD shoelaceX2(vector<Pt>& convHull) {
        TD ret = 0;
        for(int i = 0, n = convHull.size(); i < n; i++)
            ret += cross(convHull[i], convHull[(i + 1) % n]);
        return ret;
    }
    vector<Pt> createConvexHull(vector<Pt>& points) {
        sort(points.begin(), points.end());
        vector<Pt> ret;
        for(int i = 0; i < points.size(); i++) {
            while(ret.size() > 1 &&
                cross(points[i], ret[ret.size() - 1], ret[ret.size() - 2]) < -EPS)
                ret.pop_back();
            ret.push_back(points[i]);
        }
        for(int i = points.size() - 2, sz = ret.size(); i >= 0; i--) {
            while(ret.size() > sz &&
                cross(points[i], ret[ret.size() - 1], ret[ret.size() - 2]) < -EPS)
                ret.pop_back();
            if(i == 0)
                break;
        }
    }
}
```

```
ret.push_back(points[i]);
}
return ret;
bool isInside(Pt pv, vector<Pt>& x) { //using winding number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first >= pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first <= pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].second >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].second <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
    for(int i = 0; i < n; ++i) {
        if(x[i].second <= pv.second) {
            if(x[i + 1].second > pv.second && cross(x[i], x[i + 1], pv) > 0)
                ++wn;
            else if(x[i + 1].second <= pv.second && cross(x[i], x[i + 1], pv) < 0)
                --wn;
        }
        x.pop_back();
        return wn != 0;
    }
}
bool isInside(Pt pv, vector<Pt>& x) { //using winding number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first >= pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first <= pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].second >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].second <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
    for(int i = 0; i < n; ++i) {
        if(x[i].second <= pv.second) {
            if(x[i + 1].second > pv.second && cross(x[i], x[i + 1], pv) > 0)
                ++wn;
            else if(x[i + 1].second <= pv.second && cross(x[i], x[i + 1], pv) < 0)
                --wn;
        }
        x.pop_back();
        return wn != 0;
    }
}
```

4.3 Closest Pair of Points

```
#define fi first
#define se second
typedef pair<int, int> pii;
struct Point {
    int x, y, id;
};
int compareX(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->x - p2->x);
}
int compareY(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->y - p2->y);
}
```

```
double dist(Point p1, Point p2) {
    return sqrt((double)(p1.x - p2.x) * (p1.x - p2.x) +
                (double)(p1.y - p2.y) * (p1.y - p2.y)
                );
}
pair<pii, double> bruteForce(Point P[], int n) {
    double min = 1e8;
    pii ret = pii(-1, -1);
    for(int i = 0; i < n; ++i)
        for(int j = i + 1; j < n; ++j)
            if(dist(P[i], P[j]) < min) {
                ret = pii(P[i].id, P[j].id);
                min = dist(P[i], P[j]);
            }
    return pair<pii, double> (ret, min);
}
pair<pii, double> getmin(pair<pii, double> x, pair<pii, double> y) {
    if(x.fi.fi == -1 && x.fi.se == -1)
        return y;
    if(y.fi.fi == -1 && y.fi.se == -1)
        return x;
    return (x.se < y.se) ? x : y;
}
pair<pii, double> stripClosest(Point strip[], int size, double d) {
    double min = d;
    pii ret = pii(-1, -1);
    qsort(strip, size, sizeof(Point), compareY);
    for(int i = 0; i < size; ++i)
        for(int j = i + 1; j < size && (strip[j].y - strip[i].y) < min; ++j)
            if(dist(strip[i], strip[j]) < min) {
                ret = pii(strip[i].id, strip[j].id);
                min = dist(strip[i], strip[j]);
            }
    return pair<pii, double>(ret, min);
}
pair<pii, double> closestUtil(Point P[], int n) {
    if(n <= 3)
        return bruteForce(P, n);
    int mid = n / 2;
    Point midPoint = P[mid];
    pair<pii, double> dl = closestUtil(P, mid);
    pair<pii, double> dr = closestUtil(P + mid, n - mid);
    pair<pii, double> d = getmin(dl, dr);
    Point strip[n];
    int j = 0;
    for(int i = 0; i < n; i++)
        if(abs(P[i].x - midPoint.x) < d.second)
            strip[j] = P[i], j++;
    return getmin(d, stripClosest(strip, j, d.second));
}
pair<pii, double> closest(Point P[], int n) {
    qsort(P, n, sizeof(Point), compareX);
    return closestUtil(P, n);
}
Point P[50005];
int main() {
    int n;
    scanf("%d", &n);
    for(int a = 0; a < n; a++) {
        scanf("%d%d", &P[a].x, &P[a].y);
        P[a].id = a;
    }
    pair<pii, double> hasil = closest(P, n);
    if(hasil.fi.fi > hasil.fi.se)
        swap(hasil.fi.fi, hasil.fi.se);
    printf("%d %d %.6lf\n", hasil.fi.fi, hasil.fi.se, hasil.se);
    return 0;
}
```

4.4 Smallest Enclosing Circle

```
// welzl's algo to find the 2d minimum enclosing circle of a set of points
// expected O(N)
// directions: remove duplicates and shuffle points, then call welzl(points)

struct Point {
    double x;
    double y;
};

struct Circle {
    double x, y, r;
    Circle() {}
    Circle(double _x, double _y, double _r): x(_x), y(_y), r(_r) {}
};

Circle trivial(const vector<Point>& r) {
    if(r.size() == 0)
        return Circle(0, 0, -1);
    else if(r.size() == 1)
        return Circle(r[0].x, r[0].y, 0);
    else if(r.size() == 2) {
        double cx = (r[0].x + r[1].x) / 2.0, cy = (r[0].y + r[1].y) / 2.0;
        double rad = hypot(r[0].x - r[1].x, r[0].y - r[1].y) / 2.0;
        return Circle(cx, cy, rad);
    } else {
        double x0 = r[0].x, x1 = r[1].x, x2 = r[2].x;
        double y0 = r[0].y, y1 = r[1].y, y2 = r[2].y;
        double d = (x0 - x2) * (y1 - y2) - (x1 - x2) * (y0 - y2);
        double cx = (((x0 - x2) * (x0 + x2) + (y0 - y2) * (y0 + y2)) / 2 *
                    (y1 - y2) - ((x1 - x2) * (x1 + x2) + (y1 - y2) * (y1 + y2)) / 2 *
                    * (y0 - y2)) / d;
        double cy = (((x1 - x2) * (x1 + x2) + (y1 - y2) * (y1 + y2)) / 2 *
                    (x0 - x2) - ((x0 - x2) * (x0 + x2) + (y0 - y2) * (y0 + y2)) / 2 *
                    * (x1 - x2)) / d;
        return Circle(cx, cy, hypot(x0 - cx, y0 - cy));
    }
}

// SHUFFLE THE POINTS FIRST!!!!!!
Circle welzl(const vector<Point>& p, int idx = 0, vector<Point> r = {}) {
    if(idx == (int) p.size() || r.size() == 3)
        return trivial(r);
    Circle d = welzl(p, idx + 1, r);
    if(hypot(p[idx].x - d.x, p[idx].y - d.y) > d.r) {
        r.push_back(p[idx]);
        d = welzl(p, idx + 1, r);
    }
    return d;
}
```

4.5 Centroid of Polygon

$$C_x = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

$$C_y = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

4.6 Pick Theorem

A: Area of a simply closed lattice polygon
B: Number of lattice points on the edges
I: Number of points in the interior
 $A = I + \frac{B}{2} - 1$

5 Graphs

5.1 Articulation Point and Bridge

```
// gr -> adj list
// vector vis, low -> initialize to -1
// int timer -> initialize to 0
void dfs(int pos, int dad = -1) {
    vis[pos] = low[pos] = timer++;
    int kids = 0;
    for(auto& i : gr[pos]) {
        if(i == dad)
            continue;
        if(vis[i] >= 0)
            low[pos] = min(low[pos], vis[i]);
        else {
            dfs(i, pos);
            low[pos] = min(low[pos], low[i]);
            if(low[i] > vis[pos])
                is_bridge(pos, i)
                if(low[i] >= vis[pos] && dad >= 0)
                    is_articulation_point(pos)
                    ++kids;
        }
    }
    if(dad == -1 && kids > 1)
        is_articulation_point(pos)
}
```

5.2 SCC and Strong Orientation

```
#define N 10020
vector<int> adj[N];
bool vis[N], ins[N];
int disc[N], low[N], gr[N];
stack<int> st;
int id, grid;
void scc(int cur, int par) {
    disc[cur] = low[cur] = ++id;
    vis[cur] = ins[cur] = 1;
    st.push(cur);
    for(int to : adj[cur]) {
        //if (to==par) continue; // ini untuk S0(scc undirected)
        if(!vis[to])
            scc(to, cur);
        if(ins[to])
            low[cur] = min(low[cur], low[to]);
    }
    if(low[cur] == disc[cur]) {
        grid++; // group id
        while(ins[cur]) {
            gr[st.tp] = grid;
            ins[st.tp] = 0;
            st.pop();
        }
    }
}
```

5.3 Dinic's Maximum Flow

```
// O(VE log(max_flow)) if scaling == 1
// O((V + E) sqrt(E)) if unit graph (turn scaling off)
// O((V + E) sqrt(V)) if bipartite matching (turn scaling off)
// indices are 0-based
template <typename T>
struct Dinic {
    struct Edge {
        int v, rev;
```

```
T cap, flow;
};

int n;
vector<vector<Edge>> g;
vector<int> lvl, it;
static constexpr T eps = (T)1e-9;
bool scaling;

Dinic(int n, bool scaling = true): n(n), g(n), lvl(n), it(n), scaling(scaling) {}

void add_edge(int u, int v, T cap, bool directed = true) {
    Edge a{v, (int)g[v].size(), cap, 0};
    Edge b{u, (int)g[u].size(), directed ? 0 : cap, 0};
    g[u].push_back(a);
    g[v].push_back(b);
}

bool bfs(int s, int t, T lim) {
    fill(lvl.begin(), lvl.end(), -1);
    queue<int> q; q.push(s);
    lvl[s] = 0;
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (auto &e : g[u]) {
            if (lvl[e.v] == -1 && e.cap - e.flow >= lim + eps) {
                lvl[e.v] = lvl[u] + 1;
                q.push(e.v);
            }
        }
    }
    return lvl[t] != -1;
}

T dfs(int u, int t, T f) {
    if (u == t || f <= eps) return f;
    for (int &i = it[u]; i < (int)g[u].size(); i++) {
        Edge &e = g[u][i];
        if (lvl[e.v] == lvl[u] + 1 && e.cap - e.flow > eps) {
            T ret = dfs(e.v, t, min(f, e.cap - e.flow));
            if (ret > eps) {
                e.flow += ret;
                g[e.v][e.rev].flow -= ret;
                return ret;
            }
        }
    }
    return 0;
}

T max_flow(int s, int t) {
    T flow = 0;
    T lim = scaling ? (T)(1 << 30) : 1;
    while (lim > eps) {
        while (bfs(s, t, lim)) {
            fill(it.begin(), it.end(), 0);
            while (true) {
                T pushed = dfs(s, t, numeric_limits<T>::max());
                if (pushed <= eps) break;
                flow += pushed;
            }
            lim /= 2;
        }
        return flow;
    }
}

vector<int> min_cut(int s) {
    vector<int> vis(n, 0);
    queue<int> q; q.push(s); vis[s] = 1;
    while (!q.empty()) {
```

```

    int u = q.front(); q.pop();
    for (auto &e : g[u])
        if (!vis[e.v] && e.cap - e.flow > eps)
            vis[e.v] = 1, q.push(e.v);
}
return vis; // vis[i] == 1 -> S-side
}

vector<vector<int>> backtrack_paths(int s, int t) {
    vector<vector<int>> paths;
    vector<int> path;
    vector<int> used(n, 0);

    function<T(int, T)> dfs_path = [&](int u, T flow) -> T {
        if (u == t) {
            paths.push_back(path);
            return flow;
        }
        used[u] = 1;
        for (auto &e : g[u]) {
            if (!used[e.v] && e.flow > eps) {
                T pushed = min(flow, e.flow);
                path.push_back(e.v);
                T ret = dfs_path(e.v, pushed);
                e.flow -= ret;
                g[e.v][e.rev].flow += ret;
                path.pop_back();
            }
        }
        used[u] = 0;
        return 0;
    };

    path.push_back(s);
    dfs_path(s, numeric_limits<T>::max());
    return paths;
}
};

```

5.4 MCMF

```

// ---[ Min Cost Max Flow (Dijkstra + potentials, with path extraction) ]---
template<class FlowT = int, class CostT = long long>
struct MCMF { // In case CE, MCMF<ll,ll> mcmf;
    static constexpr CostT INF = numeric_limits<CostT>::max() / 4;
    static constexpr CostT EPS = (CostT)1e-9;

    struct E {
        int to, rev;
        FlowT cap, flow;
        CostT cost;
    };

    int n, s, t;
    vector<vector<E>> g;
    vector<CostT> dist, pot;
    vector<int> pv, pe;

    MCMF(int N, int S, int T) : n(N), s(S), t(T),
        g(N), dist(N), pot(N), pv(N), pe(N) {}

    void addEdge(int u, int v, FlowT cap, CostT cost) {
        E a{v, (int)g[v].size(), cap, 0, cost};
        E b{u, (int)g[u].size(), 0, 0, -cost};
        g[u].push_back(a); g[v].push_back(b);
    }

    // optional: initialize potentials if negative costs exist
    bool initPot() {
        fill(pot.begin(), pot.end(), INF);
    }
};

```

```

    pot[s] = 0;
    for (int it = 0; it < n - 1; ++it) {
        bool any = false;
        for (int u = 0; u < n; ++u) if (pot[u] < INF/2) {
            for (auto &e : g[u]) if (e.cap > 0)
                if (pot[e.to] > pot[u] + e.cost) {
                    pot[e.to] = pot[u] + e.cost;
                    any = true;
                }
            if (!any) break;
        }
        // check negative cycle
        for (int u = 0; u < n; ++u) if (pot[u] < INF/2)
            for (auto &e : g[u]) if (e.cap > 0)
                if (pot[e.to] > pot[u] + e.cost) return false;
        return true;
    }

    bool dijkstra() {
        fill(dist.begin(), dist.end(), INF);
        priority_queue<pair<CostT, int>, vector<pair<CostT, int>>, greater<>> pq;
        dist[s] = 0; pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u] + EPS) continue;
            for (int i = 0; i < (int)g[u].size(); ++i) {
                E &e = g[u][i];
                if (e.cap > e.flow) {
                    CostT nd = d + e.cost + pot[u] - pot[e.to];
                    if (nd + EPS < dist[e.to]) {
                        dist[e.to] = nd; pv[e.to] = u; pe[e.to] = i;
                        pq.push({nd, e.to});
                    }
                }
            }
        }
        return dist[t] < INF/2;
    }

    pair<FlowT, CostT> flow() {
        FlowT totF = 0; CostT totC = 0;
        while (dijkstra()) {
            for (int i = 0; i < n; ++i) if (dist[i] < INF/2) pot[i] += dist[i];
            FlowT add = numeric_limits<FlowT>::max();
            for (int v = t; v != s; v = pv[v])
                add = min(add, g[pv[v]][pe[v]].cap - g[pv[v]][pe[v]].flow);
            for (int v = t; v != s; v = pv[v]) {
                E &e = g[pv[v]][pe[v]];
                e.flow += add; g[v][e.rev].flow -= add;
                totC += (CostT)add * e.cost;
            }
            totF += add;
        }
        return {totF, totC};
    }

    // ---- backtrack flow decomposition ----
    FlowT dfsPath(int u, FlowT f, vector<int> &path, vector<vector<E>> &res) const {
        if (u == t) return f;
        for (auto &e : res[u]) if (e.flow > EPS) {
            FlowT pushed = dfsPath(e.to, min(f, e.flow), path, res);
            if (pushed > EPS) {
                e.flow -= pushed;
                path.push_back(u);
                return pushed;
            }
        }
        return 0;
    }
};

```

```
vector<pair<vector<int>,FlowT>> getPaths() const {
    auto res = g;
    vector<pair<vector<int>,FlowT>> out;
    while(true){
        vector<int> path;
        FlowT pushed = dfsPath(s, numeric_limits<FlowT>::max(), path, res);
        if(pushed <= EPS) break;
        reverse(path.begin(), path.end());
        path.push_back(t);
        out.push_back({path, pushed});
    }
    return out;
}
};
```

5.5 Flows with Demands

let S_0 be the source and T_0 be the original sink

1. add 2 additional nodes, call them S_1 and T_1
2. connect S_0 to nodes normally
3. connect nodes to T_0 normally
4. for each edge (U, V) , $\text{cap} = \text{original cap} - \text{demand}$
5. for each node N :
 1. add an edge (S_1, N) , $\text{cap} = \text{sum of inward demand to } N$
 2. add an edge (N, T_1) , $\text{cap} = \text{sum of outward demand from } N$
6. add an edge (T_0, S_0) , $\text{cap} = \text{INF}$
7. the above is not a typo!
8. run max flow normally
9. for each edge (S_1, V) and (U, T_1) , check if $\text{flow} == \text{cap}$

if step #9 fails, then it is not possible to satisfy the given demand

Mathematically, let $d(e)$ be the demand of edge e . Let V be the set of every vertex in the graph.

- $c'(S_1, v) = \sum_{u \in V} d(u, v)$ for each edge (s', v) .
- $c'(v, T_1) = \sum_{v \in V} d(v, w)$ for each edge (v, t') .
- $c'(u, v) = c(u, v) - d(u, v)$ for each edge (u, v) in the old network.
- $c'(T_0, S_0) = \infty$

5.6 Hungarian

```
template <typename TD> struct Hungarian {
    TD INF = 1e9; //max_inf
    int n;
    vector<vector<TD> > adj; // cost[left][right]
    vector<TD> hl, hr, slk;
    vector<int> fl, fr, vl, vr, pre;
    deque<int> q;
    Hungarian(int _n) {
        n = _n;
        adj = vector<vector<TD> >(n, vector<TD>(n, 0));
    }
    int check(int i) {
        if(vl[i] == 1, fl[i] != -1)
            return q.push_back(fl[i]), vr[fl[i]] = 1;
        while(i != -1)
            swap(i, fr[fl[i] = pre[i]]);
        return 0;
    }
    void bfs(int s) {
        slk.assign(n, INF);
        vl.assign(n, 0);
        vr = vl;
        q.assign(vr[s] == 1, s);
        for(TD d; ) {
            for(; !q.empty(); q.pop_front()) {
                for(int i = 0, j = q.front(); i < n; i++) {
```

```
                    if(d == hl[i] + hr[j] - adj[i][j], !vl[i] && d <= slk[i]) {
                        if(pre[i] == j, d)
                            slk[i] = d;
                        else if(!check(i))
                            return;
                    }
                }
            }
            d = INF;
            for(int i = 0; i < n; i++) if(!vl[i] && d > slk[i])
                d = slk[i];
            for(int i = 0; i < n; i++) {
                if(vl[i])
                    hl[i] += d;
                else
                    slk[i] -= d;
                if(vr[i])
                    hr[i] -= d;
            }
            for(int i = 0; i < n; i++) if(!vl[i] && !slk[i] && !check(i))
                return;
        }
    }
    TD solve() {
        fl.assign(n, -1);
        fr = fl;
        hl.assign(n, 0);
        hr = hl;
        pre.assign(n, 0);
        for(int i = 0; i < n; i++)
            hl[i] = *max_element(adj[i].begin(), adj[i].begin() + n);
        for(int i = 0; i < n; i++)
            bfs(i);
        TD ret = 0;
        for(int i = 0; i < n; i++) if(adj[i][fl[i]])
            ret += adj[i][fl[i]];
        return ret;
    }
}; //i will be matched with fl[i]
```

5.7 Edmonds' Blossom

```
// Maximum matching on general graphs in  $O(V^2 E)$ 
// Indices are 1-based
// Stolen from ko_osaga's cheatsheet
struct Blossom {
    vector<int> vis, dad, orig, match, aux;
    vector<vector<int>> conn;
    int t, N;
    queue<int> Q;

    void augment(int u, int v) {
        int pv = v;
        do {
            pv = dad[pv];
            int nv = match[pv];
            match[pv] = pv;
            match[pv] = v;
            v = nv;
        } while(u != pv);
    }

    int lca(int v, int w) {
        ++t;
        while(true) {
            if(v) {
                if(aux[v] == t)
                    return v;
                aux[v] = t;
                v = orig[dad[match[v]]];
            }
```



```

    }
    swap(v, w);
}
}

void blossom(int v, int w, int a) {
    while(orig[v] != a) {
        dad[v] = w;
        w = match[v];
        if(vis[w] == 1) {
            Q.push(w);
            vis[w] = 0;
        }
        orig[v] = orig[w] = a;
        v = dad[w];
    }
}

bool bfs(int u) {
    fill(vis.begin(), vis.end(), -1);
    iota(orig.begin(), orig.end(), 0);
    Q = queue<int>();
    Q.push(u);
    vis[u] = 0;
    while(!Q.empty()) {
        int v = Q.front();
        Q.pop();
        for(int x : conn[v]) {
            if(vis[x] == -1) {
                dad[x] = v;
                vis[x] = 1;
                if(!match[x]) {
                    augment(u, x);
                    return 1;
                }
                Q.push(match[x]);
                vis[match[x]] = 0;
            } else if(vis[x] == 0 && orig[v] != orig[x]) {
                int a = lca(orig[v], orig[x]);
                blossom(x, v, a);
                blossom(v, x, a);
            }
        }
    }
    return false;
}

Blossom(int n) : // n = vertices
    vis(n + 1), dad(n + 1), orig(n + 1), match(n + 1),
    aux(n + 1), conn(n + 1), t(0), N(n) {
    for(int i = 0; i <= n; ++i) {
        conn[i].clear();
        match[i] = aux[i] = dad[i] = 0;
    }
}

void add_edge(int u, int v) {
    conn[u].push_back(v);
    conn[v].push_back(u);
}

int solve() { // call this for answer
    int ans = 0;
    vector<int> V(N - 1);
    iota(V.begin(), V.end(), 1);
    shuffle(V.begin(), V.end(), mt19937(0x94949));
    for(auto x : V) {
        if(!match[x]) {
            for(auto y : conn[x]) {
                if(!match[y]) {
                    match[x] = y, match[y] = x;

```

```

        ++ans;
        break;
    }
}
}
}
for(int i = 1; i <= N; ++i) {
    if(!match[i] && bfs(i))
        ++ans;
}
return ans;
}
};

```

5.8 Eulerian Path or Cycle

```

// finds a eulerian path / cycle
// visits each edge only once
// properties:
// - cycle: degrees are even
// - path: degrees are even OR degrees are even except for 2 vertices
// how to use: g = adjacency list g[n] = connected to n, undirected
// if there is a vertex u with an odd degree, call dfs(u)
// else call on any vertex
// ans = path result

vector<set<int>> g;
vector<int> ans;

void dfs(int u) {
    while(g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        g[v].erase(u);
        dfs(v);
    }
    ans.push_back(u);
}

```

5.9 Hierholzer's Algorithm

```

// Eulerian on Directed Graph
stack<int> path;
vector<int> euler;
inline void hierholzer() {
    path.push(0);
    int cur = 0;
    while(!path.empty()) {
        if(!adj[cur].empty()) {
            path.push(cur);
            int next = adj[cur].back();
            adj[cur].pop();
            cur = next;
        } else {
            euler.pb(cur);
            cur = path.top();
            path.pop();
        }
    }
    reverse(euler.begin(), euler.end());
}

```

5.10 2-SAT

```

struct TwoSAT {
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, topological_order, answer;

```



```
vector<bool> vis;

TwoSAT() {}
TwoSAT(int _n) :
    n(_n), g(2 * n), gr(2 * n), comp(2 * n), answer(2 * n), vis(2 * n) {}

void add_edge(int u, int v) {
    g[u].push_back(v);
    gr[v].push_back(u);
}

// For the following three functions
// int x, bool val: if 'val' is true, we take the variable to be x.
// Otherwise we take it to be x's complement.

// At least one of them is true
void add_clause_or(int i, bool f, int j, bool p) {
    add_edge(i + (f ? n : 0), j + (p ? 0 : n));
    add_edge(j + (p ? n : 0), i + (f ? 0 : n));
}

// Only one of them is true
void add_clause_xor(int i, bool f, int j, bool p) {
    add_clause_or(i, f, j, p);
    add_clause_or(i, !f, j, !p);
}

// Both of them have the same value
void add_clause_and(int i, bool f, int j, bool p) {
    add_clause_xor(i, !f, j, p);
}

// Topological sort
void dfs(int u) {
    vis[u] = true;
    for(const auto& v : g[u])
        if(!vis[v])
            dfs(v);
    topological_order.push_back(u);
}

// Extracting strongly connected components
void scc(int u, int id) {
    vis[u] = true;
    comp[u] = id;
    for(const auto& v : gr[u])
        if(!vis[v])
            scc(v, id);
}

bool satisfiable() {
    fill(vis.begin(), vis.end(), false);
    for(int i = 0; i < 2 * n; i++)
        if(!vis[i])
            dfs(i);
    fill(vis.begin(), vis.end(), false);
    reverse(topological_order.begin(), topological_order.end());
    int id = 0;
    for(const auto& v : topological_order)
        if(!vis[v])
            scc(v, id++);
    // Constructing the answer
    for(int i = 0; i < n; i++) {
        if(comp[i] == comp[i + n])
            return false;
        answer[i] = (comp[i] > comp[i + n] ? 1 : 0);
    }
    return true;
}
};
```

6 Math

6.1 Extended Euclidean GCD

```
// computes x and y such that ax + by = gcd(a, b) in O(log (min(a, b)))
// returns {gcd(a, b), x, y}
tuple<int, int, int> gcd(int a, int b) {
    if(b == 0) return {a, 1, 0};
    auto [d, x1, y1] = gcd(b, a % b);
    return {d, y1, x1 - y1 * (a / b)};
}
```

6.2 Generalized CRT

```
template<typename T>
T extended_euclid(T a, T b, T& x, T& y) {
    if(b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    T xx, yy, gcd;
    gcd = extended_euclid(b, a % b, xx, yy);
    x = yy;
    y = xx - (yy * (a / b));
    return gcd;
}

template<typename T>
T MOD(T a, T b) {
    return (a % b + b) % b;
}

// return x, lcm. x = a % n && x = b % m
template<typename T>
pair<T, T> CRT(T a, T n, T b, T m) {
    T _n, _m;
    T gcd = extended_euclid(n, m, _n, _m);
    if(n == m) {
        if(a == b)
            return pair<T, T>(a, n);
        else
            return pair<T, T>(-1, -1);
    } else if(abs(a - b) % gcd != 0)
        return pair<T, T>(-1, -1);
    else {
        T lcm = m * n / gcd;
        T x = MOD(a + MOD(n * MOD(_n * ((b - a) / gcd), m / gcd), lcm), lcm);
        return pair<T, T>(x, lcm);
    }
}
```

6.3 Generalized Lucas Theorem

```
/*Special Lucas : (n,k) % p^x
fctp[n] = Product of the integers less than or equal
to n that are not divisible by p
Precompute fctp*/
LL p
LL E(LL n, int m) {
    LL tot = 0;
    while(n != 0)
        tot += n / m, n /= m;
    return tot;
}

LL funct(LL n, LL base) {
    LL ans = fast(fctp[base], n / base, base) * fctp[n % base] % base;
    return ans;
}
```

```
LL F(LL n, LL base) {
    LL ans = 1;
    while(n != 0) {
        ans = (ans * funct(n, base)) % base;
        n /= p;
    }
    return ans;
}
LL special_lucas(LL n, LL r, LL base) {
    p = fprime(base);
    LL pow = E(n, p) - E(n - r, p) - E(r, p);
    LL TOP = fast(p, pow, base) * F(n, base) % base;
    LL BOT = F(r, base) * F(n - r, base) % base;
    return (TOP * fast(BOT, totien(base) - 1, base)) % base;
}
//End of Special Lucas
```

6.4 Linear Diophantine

```
//FOR SOLVING MINIMUM ABS(X) + ABS(Y)
ll x, y, newX, newY, target = 0;
ll extGcd(ll a, ll b) {
    if(b == 0) {
        x = 1, y = 0;
        return a;
    }
    ll ret = extGcd(b, a % b);
    newX = y;
    newY = x - y * (a / b);
    x = newX;
    y = newY;
    return ret;
}
ll fix(ll sol, ll rt) {
    ll ret = 0;
    //CASE SOLUTION(X/Y) < TARGET
    if(sol < target)
        ret = -floor(abs(sol + target) / (double)rt);
    //CASE SOLUTION(X/Y) > TARGET
    if(sol > target)
        ret = ceil(abs(sol - target) / (double)rt);
    return ret;
}
ll work(ll a, ll b, ll c) {
    ll gcd = extGcd(a, b);
    ll solX = x * (c / gcd);
    ll solY = y * (c / gcd);
    a /= gcd;
    b /= gcd;
    ll fi = abs(fix(solX, b));
    ll se = abs(fix(solY, a));
    ll lo = min(fi, se);
    ll hi = max(fi, se);
    ll ans = abs(solX) + abs(solY);
    for(ll i = lo; i <= hi; i++) {
        ans = min(ans, abs(solX + i * b) + abs(solY - i * a));
        ans = min(ans, abs(solX - i * b) + abs(solY + i * a));
    }
    return ans;
}
```

6.5 Modular Linear Equation

```
// finds all solutions to ax = b (mod n)
vi modular_linear_equation_solver(int a, int b, int n) {
    int x, y;
    vi ret;
    int g = extended_euclid(a, n, x, y);
    if(!(b % g)) {
```

```
        x = mod(x * (b / g), n);
        for(int i = 0; i < g; i++)
            ret.push_back(mod(x + i * (n / g), n));
    }
    return ret;
}
```

6.6 Miller-Rabin and Pollard's Rho

```
namespace MillerRabin {
    const vector<ll> primes = { // deterministic up to 2^64 - 1
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37
    };
    ll gcd(ll a, ll b) {
        return b ? gcd(b, a % b) : a;
    }
    ll powa(ll x, ll y, ll p) { // (x ^ y) % p
        if(!y)
            return 1;
        if(y & 1)
            return ((__int128) x * powa(x, y - 1, p)) % p;
        ll temp = powa(x, y >> 1, p);
        return ((__int128) temp * temp) % p;
    }
    bool miller_rabin(ll n, ll a, ll d, int s) {
        ll x = powa(a, d, n);
        if(x == 1 || x == n - 1)
            return 0;
        for(int i = 0; i < s; ++i) {
            x = ((__int128) x * x) % n;
            if(x == n - 1)
                return 0;
        }
        return 1;
    }
    bool is_prime(ll x) { // use this
        if(x < 2)
            return 0;
        int r = 0;
        ll d = x - 1;
        while((d & 1) == 0) {
            d >>= 1;
            ++r;
        }
        for(auto& i : primes) {
            if(x == i)
                return 1;
            if(miller_rabin(x, i, d, r))
                return 0;
        }
        return 1;
    }
}

namespace PollardRho {
    mt19937_64 generator(chrono::steady_clock::now()
        .time_since_epoch().count());
    uniform_int_distribution<ll> rand_ll(0, LLONG_MAX);
    ll f(ll x, ll b, ll n) { // (x^2 + b) % n
        return ((__int128) x * x) % n + b) % n;
    }
    ll rho(ll n) {
        if(n % 2 == 0)
            return 2;
        ll b = rand_ll(generator);
        ll x = rand_ll(generator);
        ll y = x;
        while(1) {
            x = f(x, b, n);
            y = f(f(y, b, n), b, n);
```

```

    ll d = MillerRabin::gcd(abs(x - y), n);
    if(d != 1)
        return d;
}
}
void pollard_rho(ll n, vector<ll>& res) {
    if(n == 1)
        return;
    if(MillerRabin::is_prime(n)) {
        res.push_back(n);
        return;
    }
    ll d = rho(n);
    pollard_rho(d, res);
    pollard_rho(n / d, res);
}
vector<ll> factorize(ll n, bool sorted = 1) { // use this
    vector<ll> res;
    pollard_rho(n, res);
    if(sorted)
        sort(res.begin(), res.end());
    return res;
}
}

```

6.7 Berlekamp-Massey

```

#include <bits/stdc++.h>
using namespace std;
#define pb push_back
typedef long long ll;
#define SZ 233333
const int MOD = 1e9 + 7; //or any prime
ll qp(ll a, ll b) {
    ll x = 1;
    a %= MOD;
    while(b) {
        if(b & 1)
            x = x * a % MOD;
        a = a * a % MOD;
        b >>= 1;
    }
    return x;
}
namespace linear_seq {
    vector<int> BM(vector<int> x) {
        //ls: (shortest) relation sequence (after filling zeroes) so far
        //cur: current relation sequence
        vector<int> ls, cur;
        //lf: the position of ls (t')
        //ld: delta of ls (v')
        int lf = -1, ld = -1;
        for(int i = 0; i < int(x.size()); ++i) {
            ll t = 0;
            //evaluate at position i
            for(int j = 0; j < int(cur.size()); ++j)
                t = (t + x[i - j - 1] * (ll)cur[j]) % MOD;
            if((t - x[i]) % MOD == 0) {
                continue; //good so far
            }
            //first non-zero position
            if(!cur.size()) {
                cur.resize(i + 1);
                lf = i;
                ld = (t - x[i]) % MOD;
                continue;
            }
            //cur=cur-c/ld*(x[i]-t)
            ll k = -(x[i] - t) * qp(ld, MOD - 2) % MOD /*1/ld*/;
            vector<int> c(i - lf - 1); //add zeroes in front

```

```

            c.pb(k);
            for(int j = 0; j < int(ls.size()); ++j)
                c.pb((-ls[j]*k % MOD));
            if(c.size() < cur.size())
                c.resize(cur.size());
            for(int j = 0; j < int(cur.size()); ++j)
                c[j] = (c[j] + cur[j]) % MOD;
            //if cur is better than ls, change ls to cur
            if(i - lf + (int)ls.size() >= (int)cur.size())
                ls = cur, lf = i, ld = (t - x[i]) % MOD;
            cur = c;
        }
        for(int i = 0; i < int(cur.size()); ++i)
            cur[i] = (cur[i] % MOD + MOD) % MOD;
        return cur;
    }
    int m; //length of recurrence
    //a: first terms
    //h: relation
    ll a[SZ], h[SZ], t_[SZ], s[SZ], t[SZ];
    //calculate p*q mod f
    void mull(ll* p, ll* q) {
        for(int i = 0; i < m + m; ++i)
            t_[i] = 0;
        for(int i = 0; i < m; ++i) if(p[i])
            for(int j = 0; j < m; ++j)
                t_[i + j] = (t_[i + j] + p[i] * q[j]) % MOD;
        for(int i = m + m - 1; i >= m; --i) if(t_[i])
            //miuns t_[i]*x^{i-m}(x^m-\sum_{j=0}^{m-1} x^{m-j-1}h_j)
            for(int j = m - 1; ~j; --j)
                t_[i - j - 1] = (t_[i - j - 1] + t_[i] * h[j]) % MOD;
        for(int i = 0; i < m; ++i)
            p[i] = t_[i];
    }
    ll calc(ll K) {
        for(int i = m; ~i; --i)
            s[i] = t[i] = 0;
        //init
        s[0] = 1;
        if(m != 1)
            t[1] = 1;
        else
            t[0] = h[0];
        //binary-exponentiation
        while(K) {
            if(K & 1)
                mull(s, t);
            mull(t, t);
            K >>= 1;
        }
        ll su = 0;
        for(int i = 0; i < m; ++i)
            su = (su + s[i] * a[i]) % MOD;
        return (su % MOD + MOD) % MOD;
    }
    int work(vector<int> x, ll n) {
        if(n < int(x.size()))
            return x[n];
        vector<int> v = BM(x);
        m = v.size();
        if(!m)
            return 0;
        for(int i = 0; i < m; ++i)
            h[i] = v[i], a[i] = x[i];
        return calc(n);
    }
}
using linear_seq::work;

const vector<int> sequence = {
    0, 2, 2, 28, 60, 836, 2766

```

```
};
int main() {
    cout << work(sequence, 7) << '\n';
}
```

6.8 Fast Fourier Transform

```
using ld = double; // change to long double if reach 10^18
using cd = complex<ld>;
const ld PI = acos(-(ld)1);

void fft(vector<cd>& a, int sign = 1) {
    int n = a.size();
    ld theta = sign * 2 * PI / n;
    for(int i = 0, j = 1; j < n - 1; j++) {
        for(int k = n >> 1; k > (i ^= k); k >>= 1);
        if(j < i)
            swap(a[i], a[j]);
    }
    for(int m = 1; (m = mh << 1) <= n; mh = m) {
        int irev = 0;
        for(int i = 0; i < n; i += m) {
            cd w = exp(cd(0, theta * irev));
            for(int k = n >> 2; k > (irev ^= k); k >>= 1);
            for(int j = i; j < mh + i; j++) {
                int k = j + mh;
                cd x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
        }
    }
    if(sign == -1) for(cd& i : a)
        i /= n;
}

vector<ll> multiply(vector<ll> const& a, vector<ll> const& b) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1;
    while(n < a.size() + b.size())
        n <<= 1;
    fa.resize(n);
    fb.resize(n);
    fft(fa);
    fft(fb);
    for(int i = 0; i < n; i++)
        fa[i] *= fb[i];
    fft(fa, -1);
    vector<ll> res(n);
    for(int i = 0; i < n; i++)
        res[i] = round(fa[i].real());
    return res;
}
```

6.9 Number Theoretic Transform

```
#pragma once

const ll mod = (119 << 23) + 1, root = 62; // = 998244353
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 << 21
// and 483 << 21 (same root). The last two are > 10^9.
typedef vector<ll> vl;
void ntt(vl &a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vl rt(2, 1);
    for (static int k = 2, s = 2; k < n; k *= 2, s++) {
        rt.resize(n);
        ll z[] = {1, modpow(root, mod >> s)};
        rep(i, k, 2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
```

```
    }
    vi rev(n);
    rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j, 0, k) {
            ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
            a[i + j + k] = ai - z + (z > ai ? mod : 0);
            ai += (ai + z >= mod ? z - mod : z);
        }
    vl conv(const vl &a, const vl &b) {
        if (a.empty() || b.empty()) return {};
        int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
            n = 1 << B;
        int inv = modpow(n, mod - 2);
        vl L(a), R(b), out(n);
        L.resize(n), R.resize(n);
        ntt(L), ntt(R);
        rep(i, 0, n)
            out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv % mod;
        ntt(out);
        return {out.begin(), out.begin() + s};
    }
}
```

6.10 Gauss-Jordan

```
// Gauss-Jordan elimination with full pivoting.
//
// Uses:
// (1) solving systems of linear equations (AX=B)
// (2) inverting matrices (AX=I)
// (3) computing determinants of square matrices
//
// Running time: O(n^3)
//
// INPUT:   a[][] = an nxn matrix
//          b[][] = an nxm matrix
//
// OUTPUT:  X      = an nxm matrix (stored in b[][])
//          A^{-1} = an nxn matrix (stored in a[][])
//          returns determinant of a[][]
const double EPS = 1e-10;

typedef vector<int> VI;
typedef double T;
typedef vector<T> VT;
typedef vector<VT> VVT;
T GaussJordan(VVT& a, VVT& b) {
    const int n = a.size();
    const int m = b[0].size();
    VI irow(n), icol(n), ipiv(n);
    T det = 1;
    for(int i = 0; i < n; i++) {
        int pj = -1, pk = -1;
        for(int j = 0; j < n; j++) if(!ipiv[j])
            for(int k = 0; k < n; k++) if(!ipiv[k])
                if(pj == -1 || fabs(a[j][k]) > fabs(a[pj][pk])) {
                    pj = j;
                    pk = k;
                }
        if(fabs(a[pj][pk]) < EPS) {
            cerr << "Matrix is singular." << endl;
            exit(0);
        }
        ipiv[pk]++;
        swap(a[pj], a[pk]);
        swap(b[pj], b[pk]);
        if(pj != pk)
            det *= -1;
```

```

irow[i] = pj;
icol[i] = pk;
T c = 1.0 / a[pk][pk];
det *= a[pk][pk];
a[pk][pk] = 1.0;
for(int p = 0; p < n; p++)
    a[pk][p] *= c;
for(int p = 0; p < m; p++)
    b[pk][p] *= c;
for(int p = 0; p < n; p++) if(p != pk) {
    c = a[p][pk];
    a[p][pk] = 0;
    for(int q = 0; q < n; q++)
        a[p][q] -= a[pk][q] * c;
    for(int q = 0; q < m; q++)
        b[p][q] -= b[pk][q] * c;
}
for(int p = n - 1; p >= 0; p--) if(irow[p] != icol[p]) {
    for(int k = 0; k < n; k++)
        swap(a[k][irow[p]], a[k][icol[p]]);
}
return det;
}
int main() {
    const int n = 4;
    const int m = 2;
    double A[n][n] = { { 1, 2, 3, 4 }, { 1, 0, 1, 0 }, { 5, 3, 2, 4 }, { 6, 1, 4, 6 } };
    double B[n][m] = { { 1, 2 }, { 4, 3 }, { 5, 6 }, { 8, 7 } };
    VVT a(n), b(n);
    for(int i = 0; i < n; i++) {
        a[i] = VT(A[i], A[i] + n);
        b[i] = VT(B[i], B[i] + m);
    }
    double det = GaussJordan(a, b);
    // expected: 60
    cout << "Determinant: " << det << endl;
    // expected: -0.233333 0.166667 0.133333 0.066667
    //              0.166667 0.166667 0.333333 -0.333333
    //              0.233333 0.833333 -0.133333 -0.066667
    //              0.05 -0.75 -0.1 0.2
    cout << "Inverse: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++)
            cout << a[i][j] << ' ';
        cout << endl;
    }
    // expected: 1.63333 1.3
    //              -0.166667 0.5
    //              2.36667 1.7
    //              -1.85 -1.35
    cout << "Solution: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cout << b[i][j] << ' ';
        cout << endl;
    }
}

```

6.11 Derangement

```

der[0] = 1;
der[1] = 0;
for(int i = 2; i <= 10; ++i)
    der[i] = (i - 1) * (der[i - 1] + der[i - 2]);

```

6.12 Bernoulli Number

$$\sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{i=0}^m \binom{m+1}{i} B_i^+ n^{m+1-i} = m! \sum_{i=0}^m \frac{B_i^+ n^{m+1-i}}{i!(m+1-i)!}$$

$$B_n^+ = 1 - \sum_{i=0}^{n-1} \binom{n}{i} \frac{B_i^+}{n-i+1}, \quad B_0^+ = 1$$

6.13 Forbenius Number

$(X * Y) - (X + Y)$ and total count is $(X - 1) * (Y - 1) / 2$

6.14 Stars and Bars with Upper Bound

$$P = (1 - X^{r_1+1}) \dots (1 - X^{r_n+1}) = \sum_i c_i X^{e_i}$$

$$Ans = \sum_i c_i \binom{N - e_i + n - 1}{n - 1}$$

6.15 Arithmetic Sequences

$$U_n = a + (n - 1)a_1 + \frac{(n-1)(n-2)}{1 \times 2} a_2 + \dots + \frac{(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \times \dots} a_r$$

$$S_n = n \times a + \frac{n(n-1)}{1 \times 2} a_1 + \frac{n(n-1)(n-2)}{1 \times 2 \times 3} a_2 + \dots + \frac{n(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \dots} a_r$$

6.16 FWHT

```

// Desc : Transform a polynom to obtain a_i * b_j * x^(i XOR j) or combinations
// Time : O(N log N) with N = 2^K
// OP => c00 c01 c10 c11 | c00 c01 c10 c11 inv
// XOR => +1 +1 +1 -1 | +1 +1 +1 -1 | div the inverse with size = n
// AND => 1 +1 0 1 | 1 -1 0 1 | no comment
// OR => 1 0 +1 1 | 1 0 -1 1 | no comment
typedef vector<long long> vec;
void FWHT(vec& a) {
    int n = a.size();
    for(int lvl = 1; 2 * lvl <= n; lvl <= 1) {
        for(int i = 0; i < n; i += 2 * lvl) {
            for(int j = 0; j < lvl; j++) { // do not forget to modulo
                long long u = a[i + j], v = a[i + lvl + j];
                a[i + j] = u + v; // c00 * u + c01 * v
                a[i + lvl + j] = u - v; // c10 * u + c11 * v
            }
        }
    }
} // you can convolve as usual

```

6.17 Basis Vector

```

int basis[d]; // basis[i] keeps the mask of the vector whose f value is i
int sz; // Current size of the basis
void insertVector(int mask) {
    for(int i = 0; i < d; i++) {
        if((mask & 1 << i) == 0) {
            continue; // continue if i != f(mask)
        }
        if(!basis[i]) { // If there is no basis vector with the i'th bit set,
            // then insert this vector into the basis
            basis[i] = mask;
            ++sz;
            return;
        }
        mask ^= basis[i]; // Otherwise subtract the basis vector from this
        // vector
    }
}

```

7 Strings

7.1 Aho-Corasick

```
const int K = 26;
struct Vertex {
    int next[K];
    bool leaf = 0;
    int p = -1, ans = 0;
    char pch;
    int link = -1,mlink = -1;
    //magic link, is the link to find the nearest leaf
    int go[K];
    Vertex(int p = -1, char ch = '$') : p(p), pch(ch) {
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    }
};
vector<Vertex> t;
int add_string(string const& s) {
    int v = 0;
    for(char ch : s) {
        int c = ch - 'a';
        if(t[v].next[c] == -1) {
            t[v].next[c] = t.size();
            t.emplace_back(v, ch);
        }
        v = t[v].next[c];
    }
    t[v].leaf = 1;
    return v;
}
int go(int v, char ch);
int get_link(int v) {
    if(t[v].link == -1) {
        if(v == 0 || t[v].p == 0)
            t[v].link = 0;
        else
            t[v].link = go(get_link(t[v].p), t[v].pch);
    }
    return t[v].link;
}
int get_mlink(int v) {
    if(t[v].mlink == -1) {
        if(v == 0 || t[v].p == 0)
            t[v].mlink = 0;
        else {
            t[v].mlink = go(get_link(t[v].p), t[v].pch);
            if(t[v].mlink && !t[t[v].mlink].leaf) {
                if(t[t[v].mlink].mlink == -1)
                    get_mlink(t[t[v].mlink]);
                t[v].mlink = t[t[v].mlink].mlink;
            }
        }
    }
    return t[v].mlink;
}
int go(int v, char ch) {
    int c = ch - 'a';
    if(t[v].go[c] == -1) {
        if(t[v].next[c] != -1)
            t[v].go[c] = t[v].next[c];
        else
            t[v].go[c] = v == 0 ? 0 : go(get_link(v), ch);
    }
    return t[v].go[c];
}
//t.pb(Vertex());
```

7.2 Eertree

```
/*
    Eertree - keep track of all palindromes and its occurrences
    This code refers to problem Longest Palindromic Substring
    https://www.spoj.com/problems/LPS/
*/
#include <bits/stdc++.h>
using namespace std;
typedef long long ll;

struct node {
    int next[26];
    int sufflink;
    int len, cnt;
};

const int N = 1e5 + 69;
int n;
string s;
node tree[N];
int idx, suff;
int ans = 0;

void init_eertree() {
    idx = suff = 2;
    tree[1].len = -1, tree[1].sufflink = 1;
    tree[2].len = 0, tree[2].sufflink = 1;
}

bool add_letter(int x) {
    int cur = suff, curlen = 0;
    int nw = s[x] - 'a';
    while(1) {
        curlen = tree[cur].len;
        if(x - curlen - 1 >= 0 && s[x - curlen - 1] == s[x])
            break;
        cur = tree[cur].sufflink;
    }
    if(tree[cur].next[nw]) {
        suff = tree[cur].next[nw];
        return 0;
    }
    tree[cur].next[nw] = suff = ++idx;
    tree[idx].len = tree[cur].len + 2;
    ans = max(ans, tree[idx].len);
    if(tree[idx].len == 1) {
        tree[idx].sufflink = 2;
        tree[idx].cnt = 1;
        return 1;
    }
    while(1) {
        cur = tree[cur].sufflink;
        curlen = tree[cur].len;
        if(x - curlen - 1 >= 0 && s[x - curlen - 1] == s[x]) {
            tree[idx].sufflink = tree[cur].next[nw];
            break;
        }
    }
    tree[idx].cnt = tree[tree[idx].sufflink].cnt + 1;
    return 1;
}

int main() {
    ios::sync_with_stdio(0);
    cin.tie(0);
    cin >> n >> s;
    init_eertree();
    for(int i = 0; i < n; i++)
        add_letter(i);
    cout << ans << '\n';
}
```

```
    return 0;
}
```

7.3 Manacher's Algorithm

```
// Computes lps array. lps[i] means the longest palindromic substring centered at i (↔
// when i is even, it is between characters. when it is odd, it is on characters)lps↔
[0] = 0; lps[1] = 1;
REP(i, 2, 2 * str.size()) {
    int l = i / 2 - lps[i] / 2;
    int r = (i - 1) / 2 + lps[i] / 2;
    while(1) { // widen
        if(l == 0 || r + 1 == str.size())
            break;
        if(str[l - 1] != str[r + 1])
            break;
        l--, r++;
    }
    lps[i] = r - l + 1;
    // jump
    if(lps[i] > 2) {
        int j = i - 1, k = i + 1; // while lps[j] inside lps[i]
        while(lps[j] - j < lps[i] - i)
            lps[k++] = lps[j--];
        lps[k] = lps[i] - (i - j); // set lps[k] to edge of lps[i]
        i = k - 1; // jump to mirror, which is k
    }
}
```

7.4 Suffix Array

```
struct SuffixArray {
    string s;
    vector<int> p, c, lcp;
    int n;
    SuffixArray(string _s) : s(_s) {
        s += '$';
        n = (int)s.size();
        p.resize(n);
        c.resize(n);
        {
            // calculate for k = 0
            vector<pair<char, int>> v(n);
            for(int i = 0; i < n; ++i)
                v[i] = make_pair(s[i], i);
            sort(all(v));
            for(int i = 0; i < n; ++i)
                p[i] = v[i].se;
            c[p[0]] = 0;
            for(int i = 1; i < n; ++i)
                c[p[i]] = c[p[i - 1]] + (v[i].fi != v[i - 1].fi);
        }
        const auto countingSort = [](vector<int>& p, vector<int>& c) -> void {
            int n = (int)p.size();
            vector<int> cnt(n), pos(n);
            for(auto& i : c)
                ++cnt[i];
            for(int i = 1; i < n; ++i)
                pos[i] = pos[i - 1] + cnt[i - 1];
            vector<int> pNew(n);
            for(auto& i : p)
                pNew[pos[c[i]]++] = i;
            p = pNew;
        };
        for(int k = 0; (1 << k) < n; ++k) {
            for(int i = 0; i < n; ++i) { // transition k -> k + 1
                // shift p[i] by 2^k to the left, so that second elements are
                // sorted
                p[i] = (p[i] + n - (1 << k)) % n;
            }
        }
    }
};
```

```
    }
    countingSort(p, c);
    vector<int> cNew(n);
    for(int i = 1; i < n; ++i) {
        pair<int, int> cur = make_pair(
            c[p[i]], c[(p[i] + (1 << k)) % n]
        );
        pair<int, int> pre = make_pair(
            c[p[i - 1]], c[(p[i - 1] + (1 << k)) % n]
        );
        cNew[p[i]] = cNew[p[i - 1]] + (cur != pre);
    }
    c = cNew;
}
// lcp[i]: longest common prefix of s[p[i]] and s[p[i-1]]
lcp.resize(n); // iterate from the longest suffix
for(int i = 0, k = 0; i + 1 < n; ++i) {
    int pi = c[i]; // rank of suffix [i..]
    int j = p[pi - 1];
    for(; s[i + k] == s[j + k]; ++k)
        ;
    lcp[pi] = k;
    k = max(k - 1, 0);
}
}
```

7.5 Suffix Automaton

```
struct state {
    int len, link;
    map<char, int>next; //use array if TLE
};

const int MAXLEN = 100005;
state st[MAXLEN * 2];
int sz, last;

void sa_init() {
    sz = last = 0;
    st[0].len = 0;
    st[0].link = -1;
    st[0].next.clear();
    ++sz;
}

void sa_extend(char c) {
    int cur = sz++;
    st[cur].len = st[last].len + 1;
    st[cur].next.clear();
    int p;
    for(p = last; p != -1 && !st[p].next.count(c); p = st[p].link)
        st[p].next[c] = cur;
    if(p == -1)
        st[cur].link = 0;
    else {
        int q = st[p].next[c];
        if(st[p].len + 1 == st[q].len)
            st[cur].link = q;
        else {
            int clone = sz++;
            st[clone].len = st[p].len + 1;
            st[clone].next = st[q].next;
            st[clone].link = st[q].link;
            for(; p != -1 && st[p].next[c] == q; p = st[p].link)
                st[p].next[c] = clone;
            st[q].link = st[cur].link = clone;
        }
    }
    last = cur;
}
```

```
// forwarding
for(int i = 0; i < m; i++) {
    while(cur >= 0 && st[cur].next.count(pa[i]) == 0) {
        cur = st[cur].link;
        if(cur != -1)
            len = st[cur].len;
    }
    if(st[cur].next.count(pa[i])) {
        len++;
        cur = st[cur].next[pa[i]];
    } else
        len = cur = 0;
}
// shortening abc -> bc
if(l == m) {
    l--;
    if(l <= st[st[cur].link].len)
        cur = st[cur].link;
}
// finding lowest and highest length
int lo = st[st[cur].link].len + 1;
int hi = st[cur].len;
//Finding number of distinct substrings
//answer = distsub(0)
LL d[MAXLEN * 2];
LL distsub(int ver) {
    LL tp = 1;
    if(d[ver])
        return d[ver];
    for(map<char, int>::iterator it = st[ver].next.begin();
        it != st[ver].next.end(); it++)
        tp += distsub(it->second);
    d[ver] = tp;
    return d[ver];
}
//Total Length of all distinct substrings
//call distsub first before call lesub
LL ans[MAXLEN * 2];
LL lesub(int ver) {
    LL tp = 0;
    if(ans[ver])
        return ans[ver];
    for(map<char, int>::iterator it = st[ver].next.begin();
        it != st[ver].next.end(); it++)
        tp += lesub(it->second) + d[it->second];
    ans[ver] = tp;
    return ans[ver];
}
//find the k-th lexicographical substring
void kthsub(int ver, int K, string& ret) {
    for(map<char, int>::iterator it = st[ver].next.begin();
        it != st[ver].next.end(); it++) {
        int v = it->second;
        if(K <= d[v]) {
            K--;
            if(K == 0) {
                ret.push_back(it->first);
                return;
            } else {
                ret.push_back(it->first);
                kthsub(v, K, ret);
                return;
            }
        } else
            K -= d[v];
    }
}
// Smallest Cyclic Shift to obtain lexicographical smallest of All possible
//in int main do this
int main() {
    string S;
```

```
sa_init();
cin >> S; //input
tp = 0;
t = S.length();
S += S;
for(int a = 0; a < S.size(); a++)
    sa_extend(S[a]);
minshift(0);
}
//the function
int tp, t;
void minshift(int ver) {
    for(map<char, int>::iterator it = st[ver].next.begin();
        it != st[ver].next.end(); it++) {
        tp++;
        if(tp == t) {
            cout << st[ver].len - t + 1 << endl;
            break;
        }
        minshift(it->second);
        break;
    }
}
//end of function
// LONGEST COMMON SUBSTRING OF TWO STRINGS
string lcs(string s, string t) {
    sa_init();
    for(int i = 0; i < (int)s.length(); ++i)
        sa_extend(s[i]);
    int v = 0, l = 0,
        best = 0, bestpos = 0;
    for(int i = 0; i < (int)t.length(); ++i) {
        while(v && ! st[v].next.count(t[i])) {
            v = st[v].link;
            l = st[v].length;
        }
        if(st[v].next.count(t[i])) {
            v = st[v].next[t[i]];
            ++l;
        }
        if(l > best)
            best = l, bestpos = i;
    }
    return t.substr(bestpos - best + 1, best);
}
```

7.6 KMP

```
auto get_kmp = [&](string S, string T) -> vector<int> {
    // S is the text, T is the pattern // ababa aba -> expected: {1 2 3 2 3}
    int N = S.size(), M = T.size();
    vector<int> lps(M), kmp(N);
    for(int i = 1, j = 0; i < M; i) {
        if(T[i] == T[j])
            lps[i++] = ++j;
        else {
            if(j)
                j = lps[j - 1];
            else
                lps[i++] = 0;
        }
    }
    for(int i = 0, j = 0; i < N; i) {
        if(S[i] == T[j]) {
            kmp[i++] = ++j;
            if(j == M)
                j = lps[j - 1];
        } else {
            if(j)
                j = lps[j - 1];
        }
    }
}
```



```
        else
            kmp[i++] = 0;
    }
}
return kmp;
};
```

7.7 Z

```
auto z_function(string s) {
    int n = s.size();
    vector<int> z(n);
    int l = 0, r = 0;
    for(int i = 1; i < n; i++) {
        if(i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if(i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```

8 OEIS

8.1 A000108 (Catalan)

Catalan numbers
f(n) = nCk(2n,n) / (n+1) = nCk(2n,n) - nCk(2n,n+1) = f(n-1) * 2*(2*n-1) / (n+1)
1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, 208012, 742900,
2674440, 9694845, 35357670, 129644790, 477638700, 1767263190, 6564120420,
24466267020, 91482563640, 343059613650, 1289904147324, 4861946401452,
18367353072152, 69533550916004, 263747951750360, 1002242216651368,
3814986502092304

8.2 A018819

Binary partition function: number of partitions of n into powers of 2
f(2m+1) = f(2m); f(2m) = f(2m-1) + f(m)
1, 1, 2, 2, 4, 4, 6, 6, 10, 10, 14, 14, 20, 20, 26, 26, 36, 36, 46, 46, 60,
60, 74, 74, 94, 94, 114, 114, 140, 140, 166, 166, 202, 202, 238, 238, 284,
284, 330, 330, 390, 390, 450, 450, 524, 524, 598, 598, 692, 692, 786, 786,
900, 900, 1014, 1014, 1154, 1154, 1294, 1294

8.3 A092098

3-Portolan numbers: number of regions formed by n-secting the angles of an equilateral triangle.
long long solve(long long n) {
 long long res = (n % 2 == 1 ? 3*n*n - 3*n + 1 : 3*n*n - 6*n + 6);
 const int bats = n/2 - 1;
 for (long long i=1; i<=bats; i++) for (long long j=1; j<=bats; j++) {
 long long num = i * (n-j) * n;
 long long denum = (n-i) * j + i * (n-j);
 res -= 6 * (num % denum == 0 && num / denum <= bats);
 } return res;
}
1, 6, 19, 30, 61, 78, 127, 150, 217, 246, 331, 366, 469, 510, 625, 678, 817,
870, 1027, 1080, 1261, 1326, 1519, 1566, 1801, 1878, 2107, 2190, 2437, 2520,
2791, 2886, 3169, 3270, 3559, 3678, 3997, 4110, 4447, 4548, 4921, 5034, 5419,
5550, 5899, 6078, 6487

8.4 A000127

Maximal number of regions obtained by joining n points around a circle by straight lines
f(n) = (n^4 - 6*n^3 + 23*n^2 - 18*n + 24) / 24
1, 2, 4, 8, 16, 31, 57, 99, 163, 256, 386, 562, 794, 1093, 1471, 1941, 2517,
3214, 4048, 5036, 6196, 7547, 9109, 10903, 12951, 15276, 17902, 20854, 24158,
27841, 31931, 36457, 41449, 46938, 52956, 59536, 66712, 74519, 82993, 92171,
102091, 112792, 124314

8.5 A001534

Number of graphs with n nodes and n edges.
0, 0, 1, 2, 6, 21, 65, 221, 771, 2769, 10250, 39243, 154658, 628635, 2632420,
11353457, 50411413, 230341716, 1082481189, 5228952960, 25945377057,
132140242356, 690238318754